Part I

Logic elements and set theory

In mathematics, a fundamental activity is to establish theorems. Establishing theorems is to make a mathematical statement and provide a proof that this statement is true. The proof that this statement is correct is the demonstration that it can be obtained by using some rules called rules of logical reasoning and this by starting from other statements assumed to be true at the outset.

1 Logic elements

A proposition (or an assertion) P is the content of a declarative sentence. It is either true or false, but cannot be both true and false at the same time. We attribute to P the value 1 when P is true and the value 0 when it is false.

Example 1

1. *The integer* 4 *is even.*
2. *Cairo is the capital of Egypt.*
3. $5 \times 2 = 11$.
4. $x + 2 = 2x$.
5. *How far is Algiers to Tripoli ?*
6. *Let us go to Annaba !*

The first three sentences are propositions but the last three ones are not.

1.1 Logic connectors (Operators)

They are tools which can be used to create from given propositions new propositions whose truth values depend on the truth values of the old ones. There are five logical connectors: not, and, or, if then, if and only if. Let P and Q be two propositions:

1. The negation of the proposition P is the proposition denoted $\overline{P}$ (it reads non P) which is true when P is false and false when P is true. Its truth table is

P	$\overline{P}$
1	0
0	1

Example 2

P	x *odd*	$x \in \mathbb{Q}$	$x \leq 0$
$\overline{P}$	x *even*	$x \notin \mathbb{Q}$	$x > 0$

2. The conjunction of the propositions P and Q, is the proposition denoted $P \wedge Q$ (it reads P and Q) which is true when both P, Q are true, and false in all other cases. Its truth table is

P	Q	$P \wedge Q$
1	1	1
1	0	0
0	1	0
0	0	0

Example 3

P	Q	$P \wedge Q$
$x \geq 0$	$x \leq 0$	$x = 0$
n *odd*	$n \leq 3$	$n \in \{1, 3\}$

3. The disjunction of the propositions P and Q, is the proposition denoted $P \vee Q$ (it reads P or Q) which is true if at least one of the two statements P, Q is true, and false when both statements are false. Its truth table is

P	Q	$P \vee Q$
1	1	1
1	0	1
0	1	1
0	0	0

Remark 4 *If we want to express an exclusive disjunction "either P or Q" (i.e. one or the other, not both at the same time), we can use the proposition:* $(P \wedge \overline{Q}) \vee (\overline{P} \wedge Q)$.

4. The conditional connective (implication) of the propositions P and Q denoted by $P \Rightarrow Q$, is the proposition which is false when P is true and Q is false, and true in all other cases, (it reads P implies Q, if P then Q, P is a sufficient condition for Q or Q is a necessary condition of P), its truth table is

P	Q	$P \Rightarrow Q$
1	1	1
1	0	0
0	1	1
0	0	1

In this statement, P is called the antecedent and Q is called the consequent.

5. The biconditional (equivalence) from P to Q, denoted $P \Leftrightarrow Q$, is defined to be the proposition $(P \Rightarrow Q) \wedge (Q \Rightarrow P)$, its truth table is

P	Q	$P \Longleftrightarrow Q$
1	1	1
1	0	0
0	1	0
0	0	1

The proposition $P \Leftrightarrow Q$ reads "P and Q are equivalent", "P is equivalent to Q", "P if and only if Q" , or "P is a necessary and sufficient condition for Q".

- A tautology is a compound statement that is always true independently of its components.
- An antilogy is a compound statement which is always false independently of its components.

Example 5 *Let P and Q be two propositions. It is enough to dress the truth table to show that:*

1. *The propositions $P \vee \overline{P}$ and $P \wedge (P \Rightarrow Q) \Longrightarrow Q$ are tautologies.*
2. *The propositions $P \wedge \overline{P}$ and $P \Longleftrightarrow \overline{P}$ are antilogies.*

Let P, Q and R be three propositions. It is enough to dress the truth table to get the following:

1. $\overline{\overline{P}} \Longleftrightarrow P$.
2. $P \wedge Q \Longleftrightarrow Q \wedge P$.
3. $P \vee Q \Longleftrightarrow Q \vee P$.
4. $P \wedge (Q \wedge R) \Longleftrightarrow (P \wedge Q) \wedge R$.
5. $P \vee (Q \vee R) \Longleftrightarrow (P \vee Q) \vee R$.
6. $P \wedge (Q \vee R) \Longleftrightarrow (P \wedge Q) \vee (P \wedge R)$.
7. $P \vee (Q \wedge R) \Longleftrightarrow (P \vee Q) \wedge (P \vee R)$.
8. $(P \Rightarrow Q) \Longleftrightarrow (\overline{P} \vee Q)$.
9. $(P \Rightarrow Q) \Longleftrightarrow (\overline{Q} \Rightarrow \overline{P})$.
10. $(\overline{P \Rightarrow Q}) \Longleftrightarrow (P \wedge \overline{Q})$.
11. $(\overline{P \wedge Q}) \Longleftrightarrow (\overline{P} \vee \overline{Q})$.

12. $(\overline{P \vee Q}) \Longleftrightarrow (\overline{P} \wedge \overline{Q})$.

We have also
$(P \Rightarrow Q) \wedge (Q \Rightarrow R) \Rightarrow (Q \Rightarrow R)$.
$(P \Longleftrightarrow Q) \wedge (Q \Longleftrightarrow R) \Rightarrow (Q \Longleftrightarrow R)$.

- An axiom is a mathematical statement supposed true (that we have not proof).
- A definition is a statement which defines a mathematical object.
- A theorem is a mathematical statement that we can prove by using another true assertions and the logical connectors.
- A corollary is a theorem that we can deduce from another theorem.
- A predicate is a mathematical statement written by using variables $x, y, z, ...$ which for every fixed values for the variables we get a proposition (then it will be true or false).

2 Some notions of set theory

2.1 Definitions

A set is any collection of objects, these objects are called elements of the set. The elements in a set can be fixed by a condition, or by a common characteristic, or by enumeration. A set is denoted by a capital letter: $A, B, C, D, E ...etc$, and the elements in a set are designated by lower-case letters: $x, y, z, ...etc$.

The notion of membership is a relation between objects and sets, where we write $x \in S$ if x is an element of a set S and we say x belongs to the set S. We write $x \notin S$ if x is not an element of S and we say x does not belong to S. In a set the order of listing a set is not important.

There are two ways to define a set: the first one is called by **extension**, it concerns finite sets, it consists to write between brackets all the elements, such as $\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$, {Nouakchott, Rabat, Algiers, Tunis, Tripoli},$\{a, b, c, d, ..., z\}$.

The second one is said by **comprehension**. Let $P(x)$ be a predicate. The set of elements verifying $P(x)$ is denoted $\{x \mid P(x)\}$. (We read it: the set of x such that $P(x)$). Note that for any set S we can write $S = \left\{ x \mid \underbrace{x \in S}_{P(x)} \right\}$ which means that every set can be defined by comprehension.

Example 6

1. *The set of alphabet of the arabic language.*
2. *The set of natural integers* $\mathbb{N}$*, the set of relative integers* $\mathbb{Z}$*, the set of rational numbers* $\mathbb{Q}$*.*

3. *The set of points in the plane belonging to the circle of center O and radius* 1.

4. *The set of all functions defined on $\mathbb{R}$ and with positive values.*

5. *The set of students of Algiers university.*

2.2 Quantifiers

Let S be a set, x a variable and $P(x)$ a predicate whose statement depends on the elements x of S.

1. "For any x element of S, $P(x)$" is denoted

$$\forall x \in S, \ P(x)$$

which is true if $P(x)$ is true for any element x of S. The symbol $\forall$ reads "for all" and is called the universal quantifier.

2. "There exists an element x of S, $P(x)$" is denoted

$$\exists x \in S, \ P(x)$$

which is true if $P(x)$ is true for at least one element x of S. The symbol $\exists$ reads "it exists" and is called the existential quantifier. When we want to specify that x is unique such that $P(x)$ true we write

$$\exists \, ! \; x \in S, \ P(x).$$

Example 7

1. *The square of a real number is positive is expressed as:* $\forall x \in \mathbb{R}, x^2 \geq 0$.

2. *The equation* $\tan x = x$ *possesses a solution in the interval is expressed as:* $\exists x_0 \in \left]-\frac{\pi}{2}, \frac{\pi}{2}\right[: \tan x_0 = x_0$.

- The negation of $\forall x \in S$, $P(x)$ is $\exists x \in S$, $\overline{P(x)}$.
- The negation of $\exists x \in S$, $P(x)$ is $\forall x \in S$, $\overline{P(x)}$.
- The negation of $\forall x \in S$, $P(x) \Rightarrow Q(x)$ is $\exists x \in S$, $P(x) \wedge \overline{Q(x)}$.
- To show that the implication $\forall x \in S$, $P(x) \Rightarrow Q(x)$ is not true, it is sufficient to find an element x of S such that $(P(x) \wedge \overline{Q(x)})$ is true.
- If $P(x)$ is false for all $x \in S$, we write $\forall x \in S$, $\overline{P(x)}$.

Let P be the statement

$$(\forall x \in \mathbb{Q}) \left(x^2 + x + 1 \geq 0\right)$$

and Q the statement

$$(\exists x \in \mathbb{Q}) \left(x^2 + x + 1 < 0\right)$$

Note that $x^2 + x + 1 = \left(x + \frac{1}{2}\right)^2 + \frac{3}{4} \geq 0, \forall x \in \mathbb{Q}$ therefore P is true, and Q is false. Consider the following proposition R:

$$(\forall x \in \mathbb{Q}) (\exists y \in \mathbb{Q}) (x + y = 0)$$

The proposition R is true. In fact, for $x_0 \in \mathbb{Q}$, $x_0 + y = 0 \Leftrightarrow y = -x_0$ therefore $\exists y_0 \in \mathbb{Q}$ such that : $x_0 + y_0 = 0$. Now let us consider the proposition S:

$$(\exists y \in \mathbb{Q}) (\forall x \in \mathbb{Q}) (x + y = 0)$$

Assume that S is true. Let $\widehat{y}$ in $\mathbb{Q}$ verifying S, then S is true $\forall x \in \mathbb{Q}$, it must be true for $x = 1$, and $x = 2$ i.e.

$$\begin{cases} 1 + \widehat{y} = 0 \\ \text{and} \\ 2 + \widehat{y} = 0 \end{cases}$$

$$\begin{cases} \widehat{y} = -1 \\ \text{and} \\ \widehat{y} = -2 \end{cases}$$

This is absurd. So S is false. We conclude that the order of the quantifiers is an essential part of a proposition, and changing the order can completely change the proposition.

- When S and T are two sets, x, y variables and $P(x, y)$ a predicate depending on x, y then: $(\exists x \in S,\ \forall y \in T,\ P(x, y)) \Longrightarrow (\forall y \in T,\ \exists x \in S,\ P(x, y))$ (it is enough to take the previous example to see that the converse is true) But the converse is not true and we have $\overline{\forall x \in S, \exists y \in T,\ P(x, y)}$ is $\exists x \in S,\ \forall y \in T,\ \overline{P(x, y)}$ and $\overline{\exists x \in S, \forall y \in T, P(x, y)}$ is $\forall x \in S,\ \exists y \in T,\ \overline{P(x, y)}$.

- The statements $(\forall x \in S,\ P(x))$ and $(\forall y \in S,\ P(y))$ are equivalent and the assertions $(\exists x \in S,\ P(x))$ and $(\exists y \in S,\ P(y))$ are also equivalent. We say that x and y are **mute**.

- $\forall x \in S, \forall\ y \in T,\ P(x, y)$ and $\forall\ y \in T,\ \forall x \in S,\ P(x, y)$ are the equivalent.

- $\exists x \in S, \exists\ y \in T,\ P(x, y)$ and $\exists y \in T,\ \exists x \in S,\ P(x, y)$ are equivalent.

- In the statement $\forall x \in S,\ P(x, y, z, ...)$, the variables $y, z, ...$ are free (we can replace them by any values).

- The statement ($\forall x \in S, P(x)$ and $Q(x)$) is equivalent to the the statement ($\forall x \in S,\ P(x)$) and ($\forall x \in S,\ P(x)$).
- We have ($\forall x \in S, P(x)$ or $\forall x \in S,\ Q(x)$) $\Longrightarrow$ ($\forall x \in S, P(x)$ or $Q(x)$) but the converse is not true.

  ($S = \mathbb{R},\ P(x) : x < 3\ Q(x) : x \geq 3$ we have $\forall x \in \mathbb{R} : x < 3$ or $x \geq 3$. But we have not $\forall x \in \mathbb{R} : x < 3$ or $\forall x \in \mathbb{R} : x \geq 3$).
- We have ($\exists x \in S, P(x)$ and $Q(x)$) $\Longrightarrow$ ($\exists x \in S,\ P(x)$) and ($\exists x \in S,\ Q(x)$). But the converse is not true.

  ($\exists x \in \mathbb{N},\ x$ even and $\exists x \in \mathbb{N},\ x$ odd) does not imply $\exists x \in \mathbb{N},\ x$ even and odd).
- The statement ($\exists x \in S, P(x)$ or $Q(x)$) is equivalent to ($\exists x \in S, P(x)$ or $\exists x \in S, Q(x)$).

2.3 Inclusion and equality

Let A and B be two sets.

- We say that A is included in B, or that A is a subset of B, if the following proposition is true.

$$\forall x \in A,\ x \in B.$$

In this case we write $A \subset B$, which reads "A is included in B "

- We say that A and B are equal if the following proposition is true:

$$(A \subset B) \wedge (B \subset A)\,.$$

  In which case we write $A = B$.

2.4 Set operations

Let A and B be two subsets of a set S.

- The collection of all elements x of S that satisfy the condition: $(x \in A) \wedge (x \in B)$ true forms a set denoted $A \cap B$. It reads "A intersection B "

$$A \cap B = \{x \in S : (x \in A) \wedge (x \in B)\}\,.$$

- The collection of all elements x of S that satisfy the condition: $(x \in A) \vee (x \in B)$ true forms a set denoted $A \cup B$. It reads "A union B "

$$A \cup B = \{x \in S : (x \in A) \vee (x \in B)\}$$

- The collection of all elements x of S that satisfy the condition: $(x \in A) \wedge (x \notin B)$ true forms a set denoted $A \backslash B$. It reads "A less B "

$$A \backslash B = \{x \in S : (x \in A) \wedge (x \notin B)\}.$$

The particular case, $S \backslash A$ is called the complementary of A in S. It is denoted C_A^S, $\overline{A}$ or A^C

The set $(A \backslash B) \cup (B \backslash A)$ is denoted $A \triangle B$, called the symmetrical difference of A and B.

- We admit the existence of a set denoted $\emptyset$ verifying the condition:

For any set S we have $\emptyset \subset S$. Such a set cannot contain elements and it is unique.The set $\emptyset$ is called the empty set.

Indeed, let $A = \{a\}$ and $B = \{b\}$ be two sets, a and b being distinct. Assume that $\emptyset$ has elements, then at least one element, denoted z is in $\emptyset$. Since $\emptyset \subset A$, so $z \in A$, necessarily $z = a$. Since $\emptyset \subset B$, so $z \in B$, necessarily $z = b$ then $a = b$. Contradiction. Then $(\forall x)(x \notin \emptyset)$. Hence $\emptyset$ is empty. Let us now show that if such a set exists, then it must be unique. Suppose there are two sets $\emptyset_1, \emptyset_2$ that verify: $(\emptyset_1 \subset S)(\forall S)$ and $(\emptyset_2 \subset S)(\forall S)$. This leads for $S = \emptyset_1$ and for $S = \emptyset_2$ to $\emptyset_1 \subset \emptyset_2$ and $\emptyset_2 \subset \emptyset_1$. We deduce then $\emptyset_1 = \emptyset_2$. We can prove directly the following:

Proposition 8 *For any sets S, F and G we have*

1. $S \cap \emptyset = \emptyset$.
2. $S \cup \emptyset = S$.
3. $S \cap S = S$.
4. $S \cup S = S$.
5. $S \backslash \emptyset = S$.
6. $S \cap F = F \cap F$.
7. $S \cup F = F \cup S$.
8. $(S \cup F) \cup G = S \cup (F \cup G)$.
9. $S \cap (F \cap G) = (S \cap F) \cap G$.
10. $S \cap (F \cup G) = (S \cap F) \cup (F \cap G)$.
11. $S \cup (F \cap G) = (S \cup F) \cap (S \cup G)$.
12. $\overline{\overline{S}} = S$.
13. $\overline{(S \cup F)} = \overline{S} \cap \overline{F}$.

14. $\overline{S \cap F} = \overline{S} \cup \overline{F}$.

- Let S be a set. The set of all subsets of S is denoted $\mathcal{P}(S)$ and called power set of S. We obviously have

$$X \in \mathcal{P}(S) \text{ if an only if } X \subset S.$$

It is clear that $\mathcal{P}(S)$ is never empty since it contains $\emptyset$ and S.

Example 9

1. *Let* $S = \emptyset$. *Then* $\mathcal{P}(\emptyset) = \{\emptyset\}, \mathcal{P}(\mathcal{P}(\emptyset)) = \{\emptyset, \{\emptyset\}\}, \mathcal{P}(\mathcal{P}(\mathcal{P}(\emptyset))) = \{\emptyset, \{\emptyset\}, \{\{\emptyset\}\}, \{\emptyset, \{\emptyset\}\}\}$.
2. *Let* $S = \{1, 2, 3\}$. *Then* $\mathcal{P}(S) = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\}$.

Partition: Two subsets A and B of a set S are said to be disjoint if $A \cap B = \emptyset$. Let S be set. A partition of S is a set whose elements are nonempty disjoint subsets of S such that the union of all these subsets is all of S.

Cartesian product: Let $E_1, E_2, ..., E_n$ be n nonempty sets. Any object of the form $(x_1, x_2, ..., x_n)$, where $x_1 \in E_1, x_2 \in E_2, ..., x_n \in E_n$ is called a n tuplet. This notion obeys to the following: $(x_1, x_2, ..., x_n) = (y_1, y_2, ..., y_n)$ if and only if $x_i = y_i$, $1 \leq i \leq n$. The Cartesian product of $E_1, E_2, ..., E_n$ denoted $E_1 \times E_2 \times \cdots \times E_n$ is defined by

$$E_1 \times E_2 \times \cdots \times E_n = \{(x_1, x_2, ..., x_n) : x_1 \in E_1, x_2 \in E_2, ..., x_n \in E_n\}.$$

If $E_i = E_j, 1 \leq i, j \leq n$, the Cartesian product $\underbrace{E \times E \times \cdots \times E}_{n \text{ times}}$ is denoted E^n.

3 Methods of proof

In mathematics, the proof of a proposition consists in deducing its truth from a set of mathematical statements (called hypotheses) that are assumed to be true at the time of proof. This process is based on logic, but generally includes elements of natural language and the use of axioms, theorems, propositions, lemmas, corollaries, etc. Note that hypotheses are generally written explicitly in the statement to be proved, which is not the case for the axioms, theorems, propositions, lemmas, corollaries... used in the course of this proof. In general to prove a proposition Q, we have to prove an implication $P \Longrightarrow Q$ where P is the hypothesis and we want to get Q true. To get a proof we have to:

i) respect the logic rules,

ii) know the different manners of proofs,

iii) have good sense, rigor.

In this section, we outline the main strategies for proving a mathematical statement.

3.1 Proof by Deduction (direct proof)

Deductive reasoning is the most common type of proof. Starting with a hypothesis, a logical line of reasoning is constructed, leading to a conclusion. This method is based on the tautology

$$P \wedge (P \Longrightarrow Q) \Longrightarrow Q.$$

The proposition P represents the set of hypotheses and Q the conclusion to be reached, given that the implication $P \Longrightarrow Q$ is true.

Example 10

1. *Let us show that if a and b are odd integers, then $a+b$ is an even integer. Assume that a and b are odd integers. Then there are integers k and k' such that $a = 2k+1$ and $b = 2k'+1$. Then we have $a+b = 2k+1+2k'+1 = 2(k+k'+1)$, which shows that $x+y$ is an even integer.*

2. *Let us show that any positive integer that is a square has an odd number of divisors. Let a be a square integer. Then among the divisors of a we have 1 and itself a. As $a = b^2$ then a admits b as a divisor. This already gives three divisors $1, a$ and b. On the other hand for any other c different from 1, a and b, divisor of a , there must be an integer d such that $a = cd$. The integer d is also divisor of a and must also be different from $1, a, b$, and c. So if there are divisors other than a and b, they must be different and exist in pairs. The total number of divisors is therefore odd.*

3.2 Proof by Contrapositive

The proof by contrapositive takes advantage of the mathematical equivalence $(P \Longrightarrow Q) \Leftrightarrow (\overline{Q} \Longrightarrow \overline{P})$. In some cases, the implication $\overline{Q} \Longrightarrow \overline{P}$ is simpler to show than $P \Longrightarrow Q$. That is, a proof by contrapositive begins by assuming that Q is false (i.e., $\overline{Q}$ is true). It then produces a series of direct implications leading to the conclusion that P is false (i.e., $\overline{P}$ is true). It follows that Q cannot be false when P is true, so $P \Rightarrow Q$.

Example 11

1. *Let $n \in \mathbb{N}$. We will show that if n^2 is even, then n is even. Indeed, we prove the contrapositive of this proposition. Assume that n is odd. Then there exists $k \in \mathbb{N}$ such that $n = 2k+1$. Thus, $n^2 = (2k+1)^2 = (2k)^2$ $+4k+1 = 2(2k^2+2k)+1$. Let $k^{'} = 2k^2+2k$. Then $k^{'}$ is odd. Thus, if n is odd, then n^2 is odd. By contrapositive, if n^2 is even, then n is even.*

2. *Let m and n be two nonzero natural numbers. Let us show that if $m \times n$ is odd, then m and n are odd. Indeed, assume that m is even then there exists $k \in \mathbb{N}$ such that $m = 2k$. Then $mn = 2kn$ which is even. The case n is even is similar. Thus if m or n is even then mn is even. By contrapositive if $m \times n$ is odd, then m and n are odd.*

3.3 Proof by Contradiction

This method is based on the mathematical equivalence $\overline{(P \Longrightarrow Q)} \Longleftrightarrow P \wedge \overline{Q}$. In this case, we assume Q to be false and P to be true; if in the course of reasoning we get a contradiction, $P \wedge \overline{Q}$ cannot be true, then $P \Longrightarrow Q$ is true, we deduce that Q is true.

Example 12

1. *Let us show that $\sqrt{p} \notin \mathbb{Q}$ for any prime p. Assume for contradiction $\sqrt{p} = \frac{p_1}{q_1}$ with $p_1, q_1 \in \mathbb{Q}$ and $\gcd(p_1, q_1) = 1$. Hence $\frac{p_1^2}{q_1^2} = p$, it means $p_1^2 = pq_1^2$ and, consequently, p_1 is divisible by p. Let $p_1 = pr$, $r \in \mathbb{N}$. Then $p_1^2 = p^2r^2 = pq_1^2$, which gives $pp^2 = q_1^2$ then q_1 is also divisible by p. This contradicts the hypothesis on p_1, q_1 being coprime. So, the assumption that $\sqrt{p}$ can be expressed as $\frac{p_1}{q_1}$ with $p_1, q_1 \in \mathbb{Q}$ is false, then $\sqrt{p} \notin \mathbb{Q}$ when p is a prime.*

2. *Let us show that there is an infinity of prime numbers. Suppose there is only a finite number of prime numbers say $p_1, p_2, ..., p_k$. Put $n = 1 + p_1 p_2 \cdots p_k$. Since any positive integer different from 1 is divisible by at least one prime number, then n is divisible by some p_i where $1 \leq i \leq k$; this is impossible, as otherwise $p_i = 1$. Consequently, there is an infinity of prime numbers.*

3.4 Proof by induction

We want to prove that $(\forall n \in \mathbb{N}, n \geq n_0, P(n))$ is true where $P(n)$ is a property depending on the nonnegative integer n. First, we prove for some positive integer k_0 that $P(k_0)$ is true. Then $\forall n \in \mathbb{N}, n \geq n_0, P(n) \Longrightarrow P(n+1)$. is true. The proof by induction consists in showing that $P(n_0)$ is true, and then assuming that if $P(n)$ is true with $n \geq n_0$ then $P(n+1)$is true.

Example 13

1. *Let us show that, for any natural number n, $n^3 - n$ is a multiple of 3. For $n = 0$. $0^3 - 0 = 0$ which is divisible by 3 then $P(0)$ is valid. Suppose $P(k)$ is valid, that is, $k^3 - k$ is a multiple of 3 and prove that $P(k+1)$ is valid, in other words, we prove that $(k+1)^3 - (k+1)$ is divisible by 3.*

$$\begin{aligned}(k+1)^3 - (k+1) &= (k+1)(k+1)^2 - k - 1 \\ &= (k+1)(k^2 + 2k + 1) - k - 1 \\ &= k^3 - k + 3k^2 + 3k\end{aligned}$$

Now, by induction hypothesis, $k^3 - k = 3q$ for some $q \in \mathbb{N}$. Thus,

$$(k+1)^3 - (k+1) = 3(q + (k^2 + 1)).$$

Then $P(k+1)$ is valid. Thus, for any $n \in \mathbb{N}$, $n^3 - n$ is a multiple of 3.

2. *Let us show that for any $n \in \mathbb{N}^*$ and for any positive real x, $(1+x)^n \geq 1+nx$. For $n=1$. $P(1)$ is true since $(1+x)^1 \geq 1+x$. Assume that $(1+x)^k \geq 1+kx$ and prove that $(1+x)^{k+1} \geq 1+(k+1)x$. Multiplying the induction hypothesis $(1+x)^k \geq 1+kx$ by $(1+x)$ we obtain*

$$\begin{aligned} (1+x)^{k+1} &\geq (1+kx)(1+x) \ (\textit{since } (1+x) > 0) \\ &= 1+(k+1)x+kx^2 \\ &\geq 1+(k+1)x \ (\ \textit{since } kx^2 \geq 0) \end{aligned}$$

Then $P(k+1)$ is valid. Thus, for any $n \in \mathbb{N}^*$ and for any positive real x, $(1+x)^n \geq 1+nx$.

3.5 Proof by counter-example

An example is not enough to prove that an assertion is true, but a counter-example is enough to prove that an assertion is false. Then to reject an assertion, it is sometimes enough to find a particular case that contradicts it. This particular case is called a counter-example.

Example 14

1. *If x and y are two real numbers such that $x^2 < y^2$ then $x < y$. This assertion is false. Counter-example: for $x = 2$ and $y = -3$ we have $2^2 < (-3)^2$ but $2 > -3$.*

2. *It is well-known that if $2^n - 1$ is prime, then n is prime. The converse of this statement is false. Counter-examples: for $n = 11$ or $n = 23$ which are primes we have*
$$2^{11} - 1 = 2047 = 23 \times 89$$
and
$$2^{23} - 1 = 8388607 = 47 \times 178481.$$